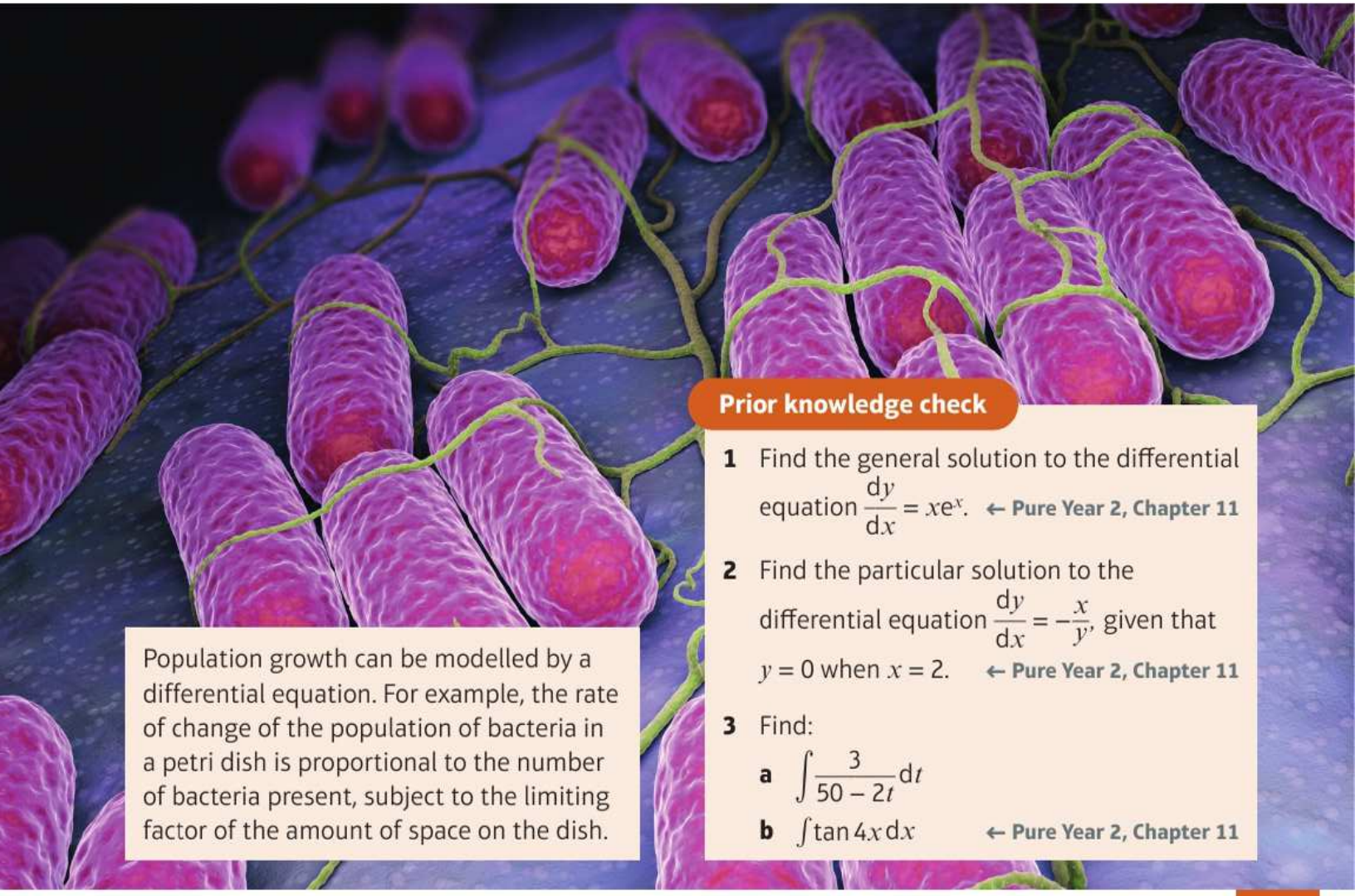
Methods in differential equations

7

Objectives

After completing this chapter you should be able to:

- Solve first-order differential equations using an integrating factor → pages 148–153
- Solve second-order homogeneous differential equations using the auxiliary equation → pages 153–157
- Solve second-order non-homogeneous differential equations using the complimentary function and the particular integral → pages 157–162
- Find particular solutions to differential equations using given boundary conditions → pages 162–165



Population growth can be modelled by a differential equation. For example, the rate of change of the population of bacteria in a petri dish is proportional to the number of bacteria present, subject to the limiting factor of the amount of space on the dish.

Prior knowledge check

1 Find the general solution to the differential equation $\frac{dy}{dx} = xe^x$. ← Pure Year 2, Chapter 11

2 Find the particular solution to the differential equation $\frac{dy}{dx} = -\frac{x}{y}$, given that $y = 0$ when $x = 2$. ← Pure Year 2, Chapter 11

3 Find:

a $\int \frac{3}{50 - 2t} dt$

b $\int \tan 4x \, dx$ ← Pure Year 2, Chapter 11

7.1 First-order differential equations

If a first-order differential equation can be written in the form $\frac{dy}{dx} = f(x)g(y)$, then you can solve it by writing $\int \frac{1}{g(y)} dy = \int f(x) dx$.

Links This process is called **separating the variables**. ← Pure Year 2, Section 11.10

Example 1

Find the general solution to the differential equation

$$\frac{dy}{dx} = -\frac{y}{x}$$

and sketch members of the family of solution curves represented by the general solution.

$\int \frac{1}{y} dy = -\int \frac{1}{x} dx$

Separate the variables and integrate.

$\ln|y| = -\ln|x| + c$

$\ln|y| + \ln|x| = c$

$\ln|xy| = c$

Collect the ln terms together and combine using the laws of logarithms.

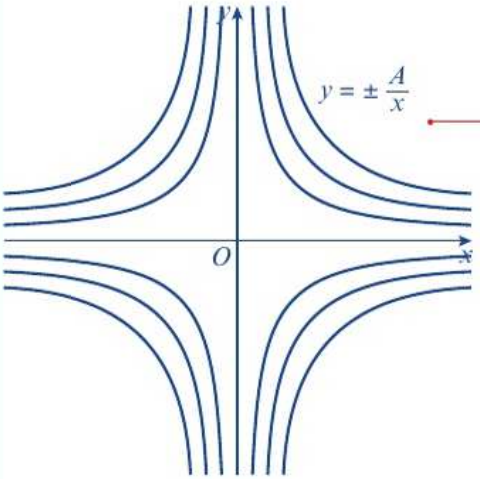
$|xy| = e^c$

$y = \pm\frac{A}{x}$, where $A = e^c$

Problem-solving

If c is a constant of integration, then $A = e^c$ can also be used as a constant. Writing the equation in this form helps you determine the family of solution curves.

Some solution curves corresponding to different values of A are shown below.



The graphs of $y = \pm\frac{A}{x}$ form the family of solution curves for this differential equation.

Online Explore families of solution curves using GeoGebra.



In this chapter, you will consider differential equations which cannot be solved by separating variables. The following example uses the product rule 'in reverse' to obtain a solution.

Example 2

Find the general solution to the differential equation

$$x^3\frac{dy}{dx} + 3x^2y = \sin x$$

$x^3\frac{dy}{dx} + 3x^2y = \sin x$

So $\frac{d}{dx}(x^3y) = \sin x$

$\Rightarrow x^3y = \int \sin x\,dx$

$= -\cos x + c$

So $y = -\frac{1}{x^3}\cos x + \frac{c}{x^3}$

You can use the product rule $u\frac{dv}{dx} + v\frac{du}{dx} = \frac{d}{dx}(uv)$, with $u = x^3$ and $v = y$, to recognise that $x^3\frac{dy}{dx} + 3x^2y = \frac{d}{dx}(x^3y)$.

Use integration as the inverse process of differentiation.

Integrate each side of the equation, including an arbitrary constant on the right-hand side.

Make y the subject by dividing each of the terms on the right-hand side by x^3.

One side of the differential equation in the example above is an exact derivative of a product in the form

$$f(x)\frac{dy}{dx} + f'(x)y = \frac{d}{dx}(f(x)y)$$

You can solve some first-order differential equations by turning them into equations of this form.

Consider the differential equation

$$\frac{dy}{dx} + \frac{3y}{x} = \frac{\sin x}{x^3}$$

The left-hand side is not a derivative of a product but if you multiply both sides by x^3 then the equation becomes the same as that in Example 2:

$$x^3\frac{dy}{dx} + 3x^2y = \sin x$$

Notation x^3 is called an **integrating factor**.

In general, if you can write a linear first-order differential equation in the form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

where $P(x)$ and $Q(x)$ are functions of x, and the function $P(x)$ can be integrated, then it is possible to find an integrating factor which will convert the left-hand side of the equation into an exact derivative of a product. Suppose that the integrating factor has the form $f(x)$, then

$$f(x)\frac{dy}{dx} + f(x)P(x)y = f(x)Q(x)$$ — Multiply the equation by the integrating factor $f(x)$.

In order for the left-hand side to be the derivative of the product $f(x)y$ it must have the form $f(x)\frac{dy}{dx} + f'(x)y$. You can see that $f(x)P(x)$ must be equal to $f'(x)$, so:

$$\int \frac{f'(x)}{f(x)}dx = \int P(x)dx$$ — Divide both sides by $f(x)$ and integrate.

$$\Rightarrow \ln|f(x)| = \int P(x)dx$$

$$\Rightarrow \quad f(x) = e^{\int P(x)dx}$$

Multiplying every term in the original differential equation by this integrating factor,

$$e^{\int P(x)dx}\frac{dy}{dx} + e^{\int P(x)dx}P(x)y = e^{\int P(x)dx}Q(x)$$

So $$\frac{d}{dx}\left(e^{\int P(x)dx}y\right) = e^{\int P(x)dx}Q(x)$$ — You can use the 'product rule in reverse' on the LHS.

$$e^{\int P(x)dx}y = \int e^{\int P(x)dx}Q(x)\,dx$$ — The equation is now directly solvable by integrating both sides.

Notice that the left-hand side is the integrating factor multiplied by y and the right-hand side is the integral of the integrating factor multiplied by $Q(x)$. This will always be the case.

- **You can solve a first-order differential equation of the form $\frac{dy}{dx} + P(x)y = Q(x)$ by multiplying every term by the integrating factor $e^{\int P(x)dx}$.**

Watch out You may write down this integrating factor without proof in your exam, but you must show all the subsequent stages of your working clearly.

Example 3

Find the general solution of the differential equation

$$\frac{dy}{dx} - 4y = e^x$$

The integrating factor is $e^{\int P(x)} = e^{\int(-4)dx} = e^{-4x}$ — Find the integrating factor.

$$e^{-4x}\frac{dy}{dx} - 4e^{-4x}y = e^x e^{-4x}$$ — Multiply each term by the integrating factor.

$$\Rightarrow \quad \frac{d}{dx}(e^{-4x}y) = e^{-3x}$$ — Express the LHS as the derivative of a product.

$$\Rightarrow \quad e^{-4x}y = \int e^{-3x}dx$$ — Integrate to get the general solution.

$$= -\tfrac{1}{3}e^{-3x} + c$$

So $$y = -\tfrac{1}{3}e^{x} + ce^{4x}$$ — Divide every term, including the constant, by the integrating factor to make y the subject.

When you have integrated and found the general solution, you can let the arbitrary constant take different numerical values, thus generating particular solutions.

In some questions you will be given a boundary condition, such as $y = 1$ when $x = 0$. You can use this to find the value of the constant. Different boundary conditions will give rise to different particular solutions.

Example 4

a Find the general solution of the differential equation $\cos x\frac{dy}{dx} + 2y\sin x = \cos^4 x$.

b Find the particular solution which satisfies the condition that $y = 2$ when $x = 0$.

a Divide through by $\cos x$:

$\frac{dy}{dx} + 2y\tan x = \cos^3 x \quad (1)$

The integrating factor is

$e^{\int P(x)dx} = e^{\int 2\tan x\,dx} = e^{2\ln\sec x} = e^{\ln\sec^2 x}$

$= \sec^2 x$

$\sec^2 x\frac{dy}{dx} + 2y\sec^2 x\tan x = \sec^2 x\cos^3 x$

So $\frac{d}{dx}(y\sec^2 x) = \cos x$

$\Rightarrow \quad y\sec^2 x = \int \cos x\,dx$

$\Rightarrow \quad y\sec^2 x = \sin x + c$

$\Rightarrow \quad y = \cos^2 x(\sin x + c)$

b $2 = \cos^2 0(\sin 0 + c)$

$\Rightarrow c = 2$

$y = \cos^2 x(\sin x + 2)$ is the required particular solution.

Divide by $\cos x$ so that equation is in the form $\frac{dy}{dx} + P(x) = Q(x)$

Use properties of ln to simplify the integrating factor.

Multiply equation (1) by the integrating factor and simplify the right-hand side.

Integrate to get the general solution and multiply through by $\cos^2 x$.

Substitute $y = 2$ and $x = 0$ into the general solution to find c.

Exercise 7A

1 For each of the following differential equations, find the general solution and sketch the family of solution curves.

a $\frac{dy}{dx} = 2x$

b $\frac{dy}{dx} = y$

c $\frac{dy}{dx} = \frac{2y}{x}$

d $\frac{dy}{dx} = \frac{x}{y}$

e $\frac{dy}{dx} = \cos x$

f $\frac{dy}{dx} = y\cot x, \quad 0 < x < \pi$

Hint You can solve all of these equations by separating variables.

2 Given that k is an arbitrary positive constant,

a show that $y^2 + kx^2 = 9k$ is the general solution to the differential equation

$\frac{dy}{dx} = \frac{-xy}{9 - x^2}, \quad |x| < 3$

b find the particular solution, which passes through the point (2, 5)

c sketch the family of solution curves for $k = \frac{1}{9}, \frac{4}{9}, 1$ and include your particular solution in the diagram.

3 Find the general solutions to these differential equations.

a $x\frac{dy}{dx} + y = \cos x$

b $e^{-x}\frac{dy}{dx} - e^{-x}y = xe^x$

c $\sin x\frac{dy}{dx} + y\cos x = 3$

d $\frac{1}{x}\frac{dy}{dx} - \frac{1}{x^2}y = e^x$

e $x^2e^y\frac{dy}{dx} + 2xe^y = x$

f $4xy\frac{dy}{dx} + 2y^2 = x^2$

E **4** **a** Find the general solution to the differential equation $\frac{dy}{dx} + 2xy = e^{-x^2}$ **(4 marks)**

b Describe the behaviour of y as $x \to \infty$. **(1 mark)**

5 a Find the general solution to the differential equation

$$x^2\frac{dy}{dx} + 2xy = 2x + 1$$

b Find the three particular solutions which pass through the points with coordinates $(-\frac{1}{2}, 0)$, $(-\frac{1}{2}, 3)$ and $(-\frac{1}{2}, 19)$ respectively and sketch their solution curves for $x < 0$.

6 a Find the general solution to the differential equation

$$\ln x\frac{dy}{dx} + \frac{y}{x} = \frac{1}{(x+1)(x+2)}, \qquad x > 1$$

b Find the particular solution which passes through the point (2, 2).

7 Find the general solutions to these differential equations by using an integrating factor.

a $\frac{dy}{dx} + 2y = e^x$

b $\frac{dy}{dx} + y\cot x = 1$

c $\frac{dy}{dx} + y\sin x = e^{\cos x}$

d $\frac{dy}{dx} - y = e^{2x}$

e $\frac{dy}{dx} + y\tan x = x\cos x$

f $\frac{dy}{dx} + \frac{y}{x} = \frac{1}{x^2}$

g $x^2\frac{dy}{dx} - xy = \frac{x^3}{x+2}, x > -2$

h $3x\frac{dy}{dx} + y = x$

i $(x+2)\frac{dy}{dx} - y = x + 2$

j $x\frac{dy}{dx} + 4y = \frac{e^x}{x^2}$

(E) **8** Find y in terms of x given that $x\frac{dy}{dx} + 2y = e^x$ and that $y = 1$ when $x = 1$. **(8 marks)**

(E) **9** Solve the differential equation, giving y in terms of x, where $x^3\frac{dy}{dx} - x^2y = 1$ and $y = 1$ at $x = 1$. **(8 marks)**

(E) **10 a** Find the general solution to the differential equation

$$\left(x + \frac{1}{x}\right)\frac{dy}{dx} + 2y = 2(x^2+1)^2$$

giving y in terms of x. **(6 marks)**

b Find the particular solution which satisfies the condition that $y = 1$ at $x = 1$. **(2 marks)**

(E) **11 a** Find the general solution to the differential equation

$$\cos x\frac{dy}{dx} + y = 1, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$$ **(6 marks)**

b Find the particular solution which satisfies the condition that $y = 2$ at $x = 0$. **(2 marks)**

(E/P) **12** Find the general solution to the differential equation

$$\frac{dy}{dx} = x\sqrt{y^2 - 4}$$ **(5 marks)**

Problem-solving

You can use the substitution $y = 2\cosh u$ to integrate $\frac{1}{\sqrt{y^2-4}}$ ← Section 6.5

E/P **13** **a** Find the general solution to the differential equation

$$\frac{dy}{dx} = y\cosh x$$ **(4 marks)**

b Find the particular solution which satisfies the condition that $y = e$ when $x = 0$. **(2 marks)**

E/P **14** **a** Find the general solution to the differential equation

$$\frac{dy}{dx} = \sqrt{1 + y^2}$$ **(4 marks)**

b Sketch the family of solution curves. **(3 marks)**

E **15** **a** Find the general solution to the differential equation

$$\cos x\frac{dy}{dx} + y\sin x = 1$$ **(6 marks)**

b Find the particular solution such that $y = 3$ when $x = \pi$. **(2 marks)**

c Show that the points $\left(\frac{\pi}{2}, 1\right)$ and $\left(\frac{3\pi}{2}, -1\right)$ lie on all possible solution curves. **(3 marks)**

E/P **16** Find a general solution to the equation $a\frac{dy}{dx} + by = 0$ in terms of a and b. **(6 marks)**

7.2 Second-order homogeneous differential equations

A second-order differential equation contains second derivatives.

Example 5

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} = 12x$$

$$\frac{dy}{dx} = 6x^2 + A$$
$$y = 2x^3 + Ax + B$$

Watch out The general solution to this second-order differential equation needs **two** arbitrary constants. If you wanted to find a particular solution you would need to know **two** boundary conditions.

In this section you will look at techniques for solving **linear** differential equations that are of the form

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$$

where a, b and c are real constants. Equations of this form (with 0 on the right-hand side) are called **second-order homogeneous** differential equations with constant coefficients.

Notation You sometimes see differential equations of this type written as $ay'' + by' + cy = 0$ where $y'' = \frac{d^2y}{dx^2}$ and $y' = \frac{dy}{dx}$

Using the techniques from the previous section, the general solution of $a\frac{dy}{dx} + by = 0$ is of the form $y = Ae^{kx}$, where $k = -\frac{b}{a}$. Notice that k is the solution to the equation $ak + b = 0$.

This suggests that an equation of the form $y = Ae^{kx}$ might also be a solution of the second-order differential equation $a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$. But it cannot be the general solution, as it only contains one arbitrary constant. Since two constants are necessary for a second-order differential equation, you can try a solution of the form $y = Ae^{\lambda x} + Be^{\mu x}$, where A and B are arbitrary constants and λ and μ are constants to be determined.

$$\frac{dy}{dx} = A\lambda e^{\lambda x} + B\mu e^{\mu x}$$
$$\frac{d^2y}{dx^2} = A\lambda^2 e^{\lambda x} + B\mu^2 e^{\mu x}$$

Substituting these into the differential equation gives

$$a(A\lambda^2 e^{\lambda x} + B\mu^2 e^{\mu x}) + b(A\lambda e^{\lambda x} + B\mu e^{\mu x}) + c(Ae^{\lambda x} + Be^{\mu x}) = 0$$
$$aA\lambda^2 e^{\lambda x} + aB\mu^2 e^{\mu x} + bA\lambda e^{\lambda x} + bB\mu e^{\mu x} + cAe^{\lambda x} + cBe^{\mu x} = 0$$
$$Ae^{\lambda x}(a\lambda^2 + b\lambda + c) + Be^{\mu x}(a\mu^2 + b\mu + c) = 0$$

This shows that the equation $y = Ae^{\lambda x} + Be^{\mu x}$ will satisfy the original differential equation if both λ and μ are solutions to the quadratic equation $am^2 + bm + c = 0$.

The equation $am^2 + bm + c = 0$ is called the **auxiliary equation.**

- **The natures of the roots α and β of the auxiliary equation, $am^2 + bm + c = 0$ determine the general solution to the differential equation $a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$.**
  You need to consider three different cases:
  - **Case 1: $b^2 > 4ac$**
    The auxiliary equation has two distinct real roots α and β. The general solution will be of the form $y = Ae^{\alpha x} + Be^{\beta x}$ where A and B are arbitrary constants.
  - **Case 2: $b^2 = 4ac$**
    The auxiliary equation has one repeated root α. The general solution will be of the form $y = (A + Bx)e^{\alpha x}$ where A and B are arbitrary constants.
  - **Case 3: $b^2 < 4ac$**
    The auxiliary equation has two complex conjugate roots α and β equal to $p \pm qi$. The general solution will be of the form $y = e^{px}(A\cos qx + B\sin qx)$ where A and B are arbitrary constants.

Links Case 3 is equivalent to $y = Ae^{\alpha x} + Be^{\beta x}$ with complex α and β. → Exercise 7B, Challenge

Note If the roots are purely imaginary ($p = 0$), the general solution reduces to $y = A\cos qx + B\sin qx$

Example 6

a Find the general solution to the equation $2\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 3y = 0$.

b Verify that your answer to part **a** satisfies the equation.

a $2m^2 + 5m + 3 = 0$ — Write down the auxiliary equation.

$(2m + 3)(m + 1) = 0 \Rightarrow m = -\frac{3}{2}$ or $m = -1$ — Solve the auxiliary equation.

So the general solution is $y = Ae^{-\frac{3}{2}x} + Be^{-x}$ where A and B are arbitrary constants — Write down the general solution.

b $y = Ae^{-\frac{3}{2}x} + Be^{-x}$

$\frac{dy}{dx} = -\frac{3}{2}Ae^{-\frac{3}{2}x} - Be^{-x}$

$\frac{d^2y}{dx^2} = \frac{9}{4}Ae^{-\frac{3}{2}x} + Be^{-x}$

$2(\frac{9}{4}Ae^{-\frac{3}{2}x} + Be^{-x}) + 5(-\frac{3}{2}Ae^{-\frac{3}{2}x} - Be^{-x}) + 3(Ae^{-\frac{3}{2}x} + Be^{-x})$

$= \frac{9}{2}Ae^{-\frac{3}{2}x} - \frac{15}{2}Ae^{-\frac{3}{2}x} + 3Ae^{-\frac{3}{2}x} + 2Be^{-x} - 5Be^{-x} + 3Be^{-x}$

$= 0$ as required.

Write down expressions for the first and second derivatives.

Substitute your expressions for $\frac{d^2y}{dx^2}$, $\frac{dy}{dx}$ and y into the differential equation, expand and simplify.

Example 7

Show that $y = (A + Bx)e^{3x}$ satisfies the differential equation $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$.

Let $y = Ae^{3x} + Bxe^{3x}$, then

$\frac{dy}{dx} = 3Ae^{3x} + 3Bxe^{3x} + Be^{3x}$

$\frac{d^2y}{dx^2} = 9Ae^{3x} + 9Bxe^{3x} + 3Be^{3x} + 3Be^{3x}$

$= 9Ae^{3x} + 9Bxe^{3x} + 6Be^{3x}$

$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 9Ae^{3x} + 9Bxe^{3x} + 6Be^{3x} - 6(3Ae^{3x} + 3Bxe^{3x} + Be^{3x}) + 9(Ae^{3x} + Bxe^{3x}) = 0$

So $y = (A + Bx)e^{3x}$ is a solution to the equation.

Differentiate the expression for y twice.

Substitute into the left-hand side of the differential equation and simplify to show that the result is zero.

Note The auxiliary equation $m^2 - 6m + 9 = 0$ has a repeated root at $m = 3$ so the general solution is in the form you expect.

Example 8

Find the general solution to the differential equation $\frac{d^2y}{dx^2} + 8\frac{dy}{dx} + 16y = 0$.

$m^2 + 8m + 16 = 0$

$(m + 4)^2 = 0 \Rightarrow m = -4$

So the general solution is $y = (A + Bx)e^{-4x}$.

Write down the auxiliary equation.

Solve the auxiliary equation. In this case there is a repeated root.

Example 9

Find the general solution to the differential equation $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 34y = 0$.

$m^2 - 6m + 34 = 0 \Rightarrow m = 3 \pm 5i$

So the general solution is

$y = e^{3x}(A\cos 5x + B\sin 5x)$

Write down the auxiliary equation and solve it using the quadratic formula or by completing the square. In this case there are two complex conjugate roots.

Example 10

Find the general solution to the differential equation $\frac{d^2y}{dx^2} + 16y = 0$.

$m^2 + 16 = 0 \Rightarrow m = \pm 4i$
So the general solution is
$y = A\cos 4x + B\sin 4x$

Write down the auxiliary equation and solve it. In this case there are two purely imaginary roots.

This is in the form $y = e^{px}(A\cos qx + B\sin qx)$ with $p = 0$.

Exercise 7B

1 Find the general solution to each to the following differential equations.

a $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$

b $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 12y = 0$

c $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 15y = 0$

d $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} - 28y = 0$

e $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} = 0$

f $3\frac{d^2y}{dx^2} + 7\frac{dy}{dx} + 2y = 0$

g $4\frac{d^2y}{dx^2} - 7\frac{dy}{dx} - 2y = 0$

h $15\frac{d^2y}{dx^2} - 7\frac{dy}{dx} - 2y = 0$

2 Find the general solution to each of the following differential equations.

a $\frac{d^2y}{dx^2} + 10\frac{dy}{dx} + 25y = 0$

b $\frac{d^2y}{dx^2} - 18\frac{dy}{dx} + 81y = 0$

c $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$

d $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 16y = 0$

e $16\frac{d^2y}{dx^2} + 8\frac{dy}{dx} + y = 0$

f $4\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + y = 0$

g $4\frac{d^2y}{dx^2} + 20\frac{dy}{dx} + 25y = 0$

h $\frac{d^2y}{dx^2} + 2\sqrt{3}\frac{dy}{dx} + 3y = 0$

3 Find the general solution to each of the following differential equations.

a $\frac{d^2y}{dx^2} + 25y = 0$

b $\frac{d^2y}{dx^2} + 81y = 0$

c $\frac{d^2y}{dx^2} + y = 0$

d $9\frac{d^2y}{dx^2} + 16y = 0$

e $\frac{d^2y}{dx^2} + 8\frac{dy}{dx} + 17y = 0$

f $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 5y = 0$

g $\frac{d^2y}{dx^2} + 20\frac{dy}{dx} + 109y = 0$

h $\frac{d^2y}{dx^2} + \sqrt{3}\frac{dy}{dx} + 3y = 0$

4 Find the general solution to each of the following differential equations.

a $\frac{d^2y}{dx^2} + 14\frac{dy}{dx} + 49y = 0$

b $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 12y = 0$

c $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 13y = 0$

d $16\frac{d^2y}{dx^2} - 24\frac{dy}{dx} + 9y = 0$

e $9\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 5y = 0$

f $6\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 0$

E/P **5** Given the differential equation $\frac{d^2x}{dt^2} + 2k\frac{dx}{dt} + 9x = 0$, where k is a real constant,

a find the general solution to the differential equation in each of the following cases:

i $|k| > 3$

ii $|k| < 3$

iii $|k| = 3$ **(8 marks)**

b In the case where $k = 2$,

i find the general solution

ii describe what happens to x as $t \to \infty$. **(4 marks)**

P **6** Given that $am^2 + bm + c = 0$ has equal roots $m = \alpha$, prove that $y = (A + Bx)e^{\alpha x}$ is a solution to the differential equation $a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + c = 0$.

P **7** Given that $y = f(x)$ and $y = g(x)$ are both solutions to the second-order differential equation $a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$, prove that $y = Af(x) + Bg(x)$, where A and B are real constants, is also a solution.

Note This result is known as the **principle of superposition**.

Challenge

Let α and β be the roots of a real-valued quadratic equation, so that $\alpha = p + iq$ and $\beta = p - iq$, $p, q \in \mathbb{R}$. Show that it is possible to choose $A, B \in \mathbb{C}$ such that $Ae^{\alpha x} + Be^{\alpha x}$ can be written in the form $e^{px}(C\cos qx + D\sin qx)$ where C and D are arbitrary real constants.

7.3 Second-order non-homogeneous differential equations

Second-order differential equations of the form

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = f(x)$$

are **non-homogeneous**.

Notation You sometimes see differential equations of this type written as $ay'' + by' + cy = f(x)$ where $y'' = \frac{d^2y}{dx^2}$ and $y' = \frac{dy}{dx}$

To solve an equation of this type you first find the general solution of the corresponding homogeneous differential equation, $a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$. This is called the **complementary function (C.F.)**.

You then need to find a **particular integral (P.I.)**, which is a function that satisfies the differential equation. The form of the particular integral depends on the form of f(x).

This table provides some particular integrals to try.

Form of f(x)	Form of particular integral
p	λ
$p + qx$	$\lambda + \mu x$
$p + qx + rx^2$	$\lambda + \mu x + \nu x^2$
pe^{kx}	λe^{kx}
$p\cos\omega x + q\sin\omega x$	$\lambda\cos\omega x + \mu\sin\omega x$

Use this form of the P.I. for functions such as $4x^2$ or $1 - x^2$.

Use this form of the P.I. for functions such as $\sin 2x$ or $5\cos x$.

- **A particular integral is a function which satisfies the original differential equation.**

Example 11

Find a particular integral of the differential equation $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = f(x)$ when f(x) is:

a 3 **b** $2x$ **c** $3x^2$ **d** e^x **e** $13\sin 3x$

a Let $y = \lambda$, then $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} = 0$

When f(x) = 3, which is constant, choose P.I. = λ, which is also constant.

Substitute into $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 3$:

$0 - 5 \times 0 + 6\lambda = 3$

Differentiate twice and substitute the derivatives into the differential equation.

$\Rightarrow \lambda = \frac{1}{2}$

Solve equation to give the value of λ.

So a particular integral is $\frac{1}{2}$

b Let $y = \lambda x + \mu$, then $\frac{dy}{dx} = \lambda$ and $\frac{d^2y}{dx^2} = 0$

When f(x) = $2x$, which is a linear function of x, choose P.I. = $\lambda x + \mu$.

Substitute into $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 2x$:

$0 - 5 \times \lambda + 6(\lambda x + \mu) = 2x$

Differentiate twice and substitute the derivatives into the differential equation.

$\Rightarrow (6\mu - 5\lambda) + 6\lambda x = 2x$

$\Rightarrow 6\mu - 5\lambda = 0$ and $6\lambda = 2$

$\Rightarrow \lambda = \frac{1}{3}$ and $\mu = \frac{5}{18}$

Equate the constant terms and the coefficients of x to give simultaneous equations, which you can solve to find λ and μ.

So a particular integral is $\frac{1}{3}x + \frac{5}{18}$

c Let $y = \lambda x^2 + \mu x + \nu$

Then $\frac{dy}{dx} = 2\lambda x + \mu$ and $\frac{d^2y}{dx^2} = 2\lambda$

As f(x) = $3x^2$, which is a quadratic function of x let P.I. = $\lambda x^2 + \mu x + \nu$.

Substitute into $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 3x^2$:

$2\lambda - 5(2\lambda x + \mu) + 6(\lambda x^2 + \mu x + \nu) = 3x^2$

$\Rightarrow (2\lambda - 5\mu + 6\nu) + (6\mu - 10\lambda)x + 6\lambda x^2 = 3x^2$

$\Rightarrow 2\lambda - 5\mu + 6\nu = 0$, $6\mu - 10\lambda = 0$ and $6\lambda = 3$

$\Rightarrow \lambda = \frac{1}{2}$, $\mu = \frac{5}{6}$ and $\nu = \frac{19}{36}$

Equate the constant terms, the coefficients of x and the coefficients of x^2 to give simultaneous equations, which you can solve to find λ, μ and ν.

So a particular integral is $\frac{1}{2}x^2 + \frac{5}{6}x + \frac{19}{36}$

d Let $y = \lambda e^x$, then $\frac{dy}{dx} = \lambda e^x$ and $\frac{d^2y}{dx^2} = \lambda e^x$

Substitute into $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^x$:

$\lambda e^x - 5\lambda e^x + 6\lambda e^x = e^x$

$\Rightarrow 2\lambda e^x = e^x$

$\Rightarrow \lambda = \frac{1}{2}$

So a particular integral is $\frac{1}{2}e^x$

As f(x) = e^x, which is an exponential function of x let P.I. = λe^x.

Equate coefficients of e^x to find the value of λ.

e Let $y = \lambda \sin 3x + \mu \cos 3x$

Then $\frac{dy}{dx} = 3\lambda \cos 3x - 3\mu \sin 3x$

and $\frac{d^2y}{dx^2} = -9\lambda \sin 3x - 9\mu \cos 3x$

Substitute into $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 13 \sin 3x$:

$-9\lambda \sin 3x - 9\mu \cos 3x - 5(3\lambda \cos 3x - 3\mu \sin 3x)$
$+ 6(\lambda \sin 3x + \mu \cos 3x) = 13 \sin 3x$

$\Rightarrow (-9\lambda + 15\mu + 6\lambda) \sin 3x +$
$(-9\mu - 15\lambda + 6\mu) \cos 3x = 13 \sin 3x$

$\Rightarrow -9\lambda + 15\mu + 6\lambda = 13$
and $-9\mu - 15\lambda + 6\mu = 0$

$\Rightarrow \lambda = -\frac{1}{6}$ and $\mu = \frac{5}{6}$

So a particular integral is $-\frac{1}{6} \sin 3x + \frac{5}{6} \cos 3x$

As f(x) = 13 sin 3x, which is a trigonometric function of x let P.I. = $\lambda \sin 3x + \mu \cos 3x$, also a similar trigonometric function.

Problem-solving

Equate coefficients of sin 3x and of cos 3x and solve simultaneous equations.

- **To find the general solution to the differential equation $a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = f(x)$,**
  - **Solve the corresponding homogeneous equation $a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$ to find the complementary function (C.F.)**
  - **Choose an appropriate form for the particular integral (P.I.) and substitute into the original equation to find the values of any coefficients.**
  - **The general solution is y = C.F. + P.I.**

Example 12

Find the general solution to the differential equation $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = f(x)$ when f(x) is:

a 3 **b** $2x$ **c** $3x^2$ **d** e^x **e** $13 \sin 3x$

$m^2 - 5m + 6 = 0$

$(m - 3)(m - 2) = 0 \Rightarrow m = 3$ or $m = 2$

Hence the complementary function is

$y = Ae^{3x} + Be^{2x}$ where A and B are arbitrary constants.

Solve the auxiliary equation to find the values of m.

The particular integrals were already found in example 11 so the general solutions are:

a $y = Ae^{3x} + Be^{2x} + \frac{1}{2}$

b $y = Ae^{3x} + Be^{2x} + \frac{1}{3}x + \frac{5}{18}$

c $y = Ae^{3x} + Be^{2x} + \frac{1}{2}x^2 + \frac{5}{6}x + \frac{19}{36}$

d $y = Ae^{3x} + Be^{2x} + \frac{1}{2}e^x$

e $y = Ae^{3x} + Be^{2x} - \frac{1}{6}\sin 3x + \frac{5}{6}\cos 3x$

The general solution is y = C.F. + P.I.

You need to be careful if the standard form of the particular integral contains terms which form part of the complementary function. If this is the case, you need to modify your particular integral so that no two terms in the general solution have the same form.

For example, this situation occurs when f(x) is of the form pe^{kx}, and k is one of the roots of the auxiliary equation. In this case you can try a particular integral of the form, λxe^{kx}.

Example 13

Find the general solution to the differential equation $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^{2x}$

As in Example 12, the complementary function is $y = Ae^{3x} + Be^{2x}$.
The particular integral cannot be λe^{2x}, as this is part of the complementary function.

So let $y = \lambda xe^{2x}$

Then $\frac{dy}{dx} = 2\lambda xe^{2x} + \lambda e^{2x}$

and

$\frac{d^2y}{dx^2} = 4\lambda xe^{2x} + 2\lambda e^{2x} + 2\lambda e^{2x} = 4\lambda xe^{2x} + 4\lambda e^{2x}$

Substitute into $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^{2x}$:

$4\lambda xe^{2x} + 4\lambda e^{2x} - 5(2\lambda xe^{2x} + \lambda e^{2x}) + 6\lambda xe^{2x} = e^{2x}$

$\Rightarrow -\lambda e^{2x} = e^{2x}$

$\Rightarrow \lambda = -1$

So a particular integral is $-xe^{2x}$.

The general solution is $y = Ae^{3x} + Be^{2x} - xe^{2x}$.

Watch out The function λe^{2x} is **part of the C.F.** and satisfies the differential equation $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$, so it cannot also satisfy $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^{2x}$.

Let the P.I. be λxe^{2x} and differentiate, substitute and solve to find λ.

The general solution is y = C.F. + P.I.

When one of the roots of the auxiliary equation is 0, the complementary function will contain a constant term. If f(x) is a polynomial, you will need to multiply its particular integral by x to make sure the P.I. does not also contain a constant term.

Example 14

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = 3$$

First consider the equation $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = 0$

$m^2 - 2m = 0$

$m(m - 2) = 0$

$\Rightarrow m = 0$ or $m = 2$

So the complementary function is $y = A + Be^{2x}$.

The particular integral cannot be a constant, as this is part of the complementary function, so let $y = \lambda x$.

Then $\frac{dy}{dx} = \lambda$ and $\frac{d^2y}{dx^2} = 0$

Substitute into $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = 3$:

$0 - 2\lambda = 3$

$\Rightarrow \lambda = -\frac{3}{2}$

So a particular integral is $-\frac{3}{2}x$

The general solution is $y = A + Be^{2x} - \frac{3}{2}x$.

Find the complementary function by putting the right-hand side of the differential equation equal to zero, and solving this new equation.

Write down and solve the auxiliary equation.

Try to find a particular integral. The right-hand side of the original equation was 3, which was a constant and usually this would imply a constant P.I.

As the C.F. includes a constant term 'A', the P.I. cannot also be constant. A value of λ would satisfy $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = 0$ rather than $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = 3$.

Multiply the 'expected' P.I. by x and try λx instead.

The general solution is y = C.F. + P.I.

Exercise 7C

1 Solve each of the following differential equations, giving the general solution.

a $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 5y = 10$

b $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 12y = 36x$

c $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 12y = 12e^{2x}$

d $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 15y = 5$

e $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 16y = 8x + 12$

f $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 25\cos 2x$

g $\frac{d^2y}{dx^2} + 81y = 15e^{3x}$

h $\frac{d^2y}{dx^2} + 4y = \sin x$

i $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 5y = 25x^2 - 7$

j $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 26y = e^x$

(E) **2 a** Find a particular integral for the differential equation

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 4y = x^2 - 3x + 2$$ **(6 marks)**

b Hence find the general solution. **(3 marks)**

E/P **3** y satisfies the differential equation

$$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} = 2x^2 - x + 1$$

a Find the complementary function for this differential equation. **(3 marks)**

b Hence find a suitable particular integral and write down the general solution to the differential equation. **(7 marks)**

Hint Try a particular integral of the form $\lambda x + \mu x^2 + \nu x^3$.

E/P **4** Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} = 24x^2$$ **(10 marks)**

E/P **5** **a** Explain why λxe^x is not a suitable form for the particular integral for the differential equation

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = e^x$$ **(2 marks)**

b Find the value of λ for which λx^2e^x is a particular integral for the differential equation. **(5 marks)**

c Hence find the general solution. **(3 marks)**

E/P **6** $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 3y = kt + 5$, where k is a constant and $t > 0$.

a Find the general solution to the differential equation in terms of k. **(7 marks)**

For large values of t, this general solution may be approximated by a linear function.

b Given that $k = 6$, find the equation of this linear function. **(2 marks)**

Challenge

Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} + y = 5xe^{2x}$$

7.4 Using boundary conditions

You can use given boundary conditions to find a particular solution to a second-order differential equation. Since there are two arbitrary constants, you will need two boundary conditions to determine the complete particular solution.

Example 15

Find y in terms of x, given that $\frac{d^2y}{dx^2} - y = 2e^x$, and that $\frac{dy}{dx} = 0$ and $y = 0$ at $x = 0$.

First consider the equation $\frac{d^2y}{dx^2} - y = 0$.

$m^2 - 1 = 0 \Rightarrow m = \pm 1$

So the complementary function is $y = Ae^x + Be^{-x}$.

Solve the auxiliary equation to find the values of m.

The particular integral cannot be λe^x, as this is part of the complementary function, so let $y = \lambda x e^x$.

Then $\frac{dy}{dx} = \lambda x e^x + \lambda e^x$ and $\frac{d^2y}{dx^2} = \lambda x e^x + \lambda e^x + \lambda e^x$

Substitute into $\frac{d^2y}{dx^2} - y = 2e^x$:

$\lambda x e^x + \lambda e^x + \lambda e^x - \lambda x e^x = 2e^x$

$\Rightarrow \lambda = 1$

So a particular integral is xe^x.

The general solution is

$y = Ae^x + Be^{-x} + xe^x$

Since $y = 0$ at $x = 0$, $0 = A + B$

$\Rightarrow \quad A + B = 0$

Differentiating $y = Ae^x + Be^{-x} + xe^x$ with respect to x gives

$\frac{dy}{dx} = Ae^x - Be^{-x} + e^x + xe^x$

Since $\frac{dy}{dx} = 0$ at $x = 0$, $0 = A - B + 1$

$\Rightarrow \quad A - B = -1$

Solving the simultaneous equations gives

$A = -\frac{1}{2}$, and $B = \frac{1}{2}$

So $y = -\frac{1}{2}e^x + \frac{1}{2}e^{-x} + xe^x$ is the required solution.

As λe^x satisfies $\frac{d^2y}{dx^2} - y = 0$, it cannot also satisfy $\frac{d^2y}{dx^2} - y = 2e^x$.

Substitute the boundary condition, $y = 0$ at $x = 0$, into the general solution to obtain an equation relating A and B.

Substitute the second boundary condition, $\frac{dy}{dx} = 0$ at $x = 0$, into the derivative of the general solution, to obtain a second equation relating A and B.

Solve the two equations to find values for A and B.

Example 16

Given that a particular integral is of the form $\lambda \sin 2t$, find the solution to the differential equation $\frac{d^2x}{dt^2} + x = 3\sin 2t$, for which $x = 0$ and $\frac{dx}{dt} = 1$ when $t = 0$.

First consider the equation $\frac{d^2x}{dt^2} + x = 0$.

$m^2 + 1 = 0 \Rightarrow m = \pm i$

So, the complementary function is

$x = A\cos t + B\sin t$.

The particular integral is $\lambda \sin 2t$, so let $x = \lambda \sin 2t$.

Then $\frac{dx}{dt} = 2\lambda \cos 2t$ and $\frac{d^2x}{dt^2} = -4\lambda \sin 2t$

Substitute into $\frac{d^2x}{dt^2} + x = 3\sin 2t$:

$-4\lambda \sin 2t + \lambda \sin 2t = 3\sin 2t$

$\Rightarrow \lambda = -1$

Solve the auxiliary equation to find the values of m.

Watch out Normally you would need to try a particular integral of the form $\lambda \sin 2t + \mu \cos 2t$ for this equation. However, in this case you are told that there is a particular integral in the form $\lambda \sin 2t$.

So a particular integral is $-\sin 2t$.

The general solution is

$x = A\cos t + B\sin t - \sin 2t$

Use general solution = complementary function + particular integral.

Since $x = 0$ at $t = 0$, $A = 0$.

Substitute the initial condition, $x = 0$ at $t = 0$, into the general solution to obtain $A = 0$.

Differentiating $x = B\sin t - \sin 2t$ with respect to t gives

$\frac{dx}{dt} = B\cos t - 2\cos 2t$

Since $\frac{dx}{dt} = 1$ at $t = 0$, $1 = B - 2$

Substitute the second initial condition, $\frac{dx}{dt} = 1$ at $t = 0$, into the derivative of the general solution, to obtain a second equation leading to $B = 3$.

$\Rightarrow \quad B = 3$

And so $x = 3\sin t - \sin 2t$ is the required solution.

Exercise 7D

E **1 a** Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 12e^x$$ **(5 marks)**

b Hence find the particular solution that satisfies $y = 1$ and $\frac{dy}{dx} = 0$ when $x = 0$. **(4 marks)**

E **2 a** Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = 12e^{2x}$$ **(5 marks)**

b Hence find the particular solution that satisfies $y = 2$ and $\frac{dy}{dx} = 6$ when $x = 0$. **(5 marks)**

E **3** Given that $y = 0$ and $\frac{dy}{dx} = \frac{1}{6}$ when $x = 0$, find the particular solution to the differential equation

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 42y = 14$$ **(10 marks)**

E **4 a** Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} + 9y = 16\sin x$$ **(6 marks)**

b Hence find the particular solution that satisfies $y = 1$ and $\frac{dy}{dx} = 8$ when $x = 0$. **(6 marks)**

E **5 a** Find the general solution to the differential equation

$$4\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 5y = \sin x + 4\cos x$$ **(6 marks)**

b Hence find the particular solution that satisfies $y = 0$ and $\frac{dy}{dx} = 0$ when $x = 0$. **(6 marks)**

E/P **6 a** Find the general solution to the differential equation

$$\frac{d^2x}{dt^2} - 3\frac{dx}{dt} + 2x = 2t - 3$$ **(6 marks)**

b Given that $x = 1$ when $t = 0$, and $x = 2$ when $t = 1$, find a particular solution of this differential equation. **(6 marks)**

E **7** Find the particular solution to the differential equation

$$\frac{d^2x}{dt^2} - 9x = 10\sin t$$

that satisfies $x = 2$ and $\frac{dx}{dt} = -1$ when $t = 0$. **(10 marks)**

E/P **8 a i** Find the value of λ for which $y = \lambda t^3 e^{2t}$ is a particular solution to the differential equation

$$\frac{d^2x}{dt^2} - 4\frac{dx}{dt} + 4x = 3te^{2t}$$

ii Hence find the general solution to the differential equation. **(6 marks)**

b Find the particular solution that satisfies $x = 0$ and $\frac{dx}{dt} = 1$ when $t = 0$. **(6 marks)**

E **9** Find the particular solution to the differential equation

$$25\frac{d^2x}{dt^2} + 36x = 18$$

that satisfies $x = 1$ and $\frac{dx}{dt} = 0.6$ when $t = 0$. **(12 marks)**

E **10 a** Find the general solution to the differential equation

$$\frac{d^2x}{dt^2} - 2\frac{dx}{dt} + 2x = 2t^2$$ **(6 marks)**

b Hence find the particular solution that satisfies $x = 1$ and $\frac{dx}{dt} = 3$ when $t = 0$. **(6 marks)**

E/P **11 a** Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 3e^{2x}$$ **(7 marks)**

b Hence find the particular solution that satisfies $y = 0$, $\frac{dy}{dx} = 0$ when $x = 0$. **(6 marks)**

E/P **12** Solve the differential equation

$$\frac{d^2y}{dx^2} + 9y = \sin 3x$$

subject to the boundary conditions $y = 0$, $\frac{dy}{dx} = 0$ when $x = 0$. **(14 marks)**

E/P **13** $\frac{d^2x}{dt^2} + 5\frac{dx}{dt} + 6x = 2e^{-t}$

Given that $x = 0$ and $\frac{dx}{dt} = 2$ at $t = 0$,

a find x in terms of t. **(8 marks)**

b Show that the maximum value of x is $\frac{2\sqrt{3}}{9}$ and justify that this is a maximum. **(7 marks)**

Mixed exercise 7

E **1** Find the general solution to the differential equation

$$\frac{dy}{dx} + y\tan x = 2\sec x$$

giving your answer in the form $y = f(x)$. **(7 marks)**

E **2** Find the general solution to the differential equation

$$(1 - x^2)\frac{dy}{dx} + xy = 5x, \quad -1 < x < 1$$

giving your answer in the form $y = f(x)$. **(7 marks)**

E **3** Find the general solution to the differential equation

$$x\frac{dy}{dx} + x + y = 0$$

giving your answer in the form $y = f(x)$. **(7 marks)**

E **4** y satisfies the differential equation

$$\frac{dy}{dx} + \frac{y}{x} = \sqrt{x}$$

Find y as a function of x. **(7 marks)**

E **5** y satisfies the differential equation

$$\frac{dy}{dx} + 2xy = x$$

Find y in terms of x. **(7 marks)**

E/P **6** Find the general solution to the differential equation

$$x(1 - x^2)\frac{dy}{dx} + (2x^2 - 1)y = 2x^3, \quad 0 < x < 1$$

giving your answer in the form $y = f(x)$. **(7 marks)**

E/P **7** Find the general solution to the equation $\frac{dy}{dx} - ay = Q(x)$, where a is a constant, giving your answer in terms of a, when

a $Q(x) = ke^{\lambda x}$ (k and λ are constants). **(6 marks)**

Given that $Q(x) = kx^n e^{ax}$, where k and n are constants,

b find the general solution to the differential equation. **(7 marks)**

E/P **8** Find, in the form $y = f(x)$, the general solution to the differential equation

$$\tan x\frac{dy}{dx} + y = 2\cos x \tan x, \ 0 < x < \frac{\pi}{2}$$ **(6 marks)**

E **9** **a** Find the general solution to the differential equation

$$\frac{dy}{dx} + y\tan x = e^x \cos x, -\frac{\pi}{2} < x < \frac{\pi}{2}$$

giving your answer in the form $y = f(x)$. **(6 marks)**

b Find the particular solution for which $y = 1$ at $x = \pi$. **(3 marks)**

E **10** $\frac{dy}{dx} - 3y = \sin x$

Given that $y = 0$ when $x = 0$, find y in terms of x. **(7 marks)**

E/P **11** **a** Find the general solution to the differential equation

$$\frac{dy}{dx} = y\sinh x$$ **(4 marks)**

b Find the particular solution which satisfies the condition that $y = 1$ when $x = 0$. **(2 marks)**

E **12** $\frac{dy}{dx} = x(4 - y^2)$

Given that $y = 1$ when $x = 0$, find y in terms of x. **(7 marks)**

E **13** Find the general solution to the differential equation $\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$ **(6 marks)**

E **14** Find the general solution to the differential equation $\frac{d^2y}{dx^2} - 12\frac{dy}{dx} + 36y = 0$ **(6 marks)**

E **15** Find the general solution to the differential equation $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} = 0$ **(6 marks)**

E/P **16** Find y in terms of k and x, given that $\frac{d^2y}{dx^2} + k^2y = 0$ where k is a constant, and $y = 1$ and $\frac{dy}{dx} = 1$ at $x = 0$. **(8 marks)**

E **17** Find the solution to the differential equation $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 10y = 0$ for which $y = 0$ and $\frac{dy}{dx} = 3$ at $x = 0$. **(8 marks)**

E **18** **a** Find the value of k for which $y = ke^{2x}$ is a particular integral of the differential equation

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = e^{2x}$$ **(4 marks)**

b Using your answer to part **a**, find the general solution to the differential equation. **(5 marks)**

E/P **19** Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} - y = 4e^x$$ **(7 marks)**

E/P **20** The differential equation $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 4e^{2x}$ is to be solved.

a Find the complementary function. **(3 marks)**

b Explain why neither λe^{2x} nor λxe^{2x} can be a particular integral for this equation. **(2 marks)**

A particular integral has the form kx^2e^{2x}.

c Determine the value of the constant k and find the general solution of the equation. **(6 marks)**

E/P **21** Find the particular solution of the differential equation

$$\frac{d^2y}{dt^2} + 4y = 5\cos 3t$$

which satisfies the initial conditions that when $t = 0$, $y = 1$ and $\frac{dy}{dt} = 2$. **(12 marks)**

(E) **22** **a** Find the values of λ, μ and k such that $y = \lambda + \mu x + kxe^{2x}$ is a particular integral of the differential equation

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 4x + e^{2x}$$ **(5 marks)**

b Using your answer to part **a**, find the general solution of the differential equation. **(5 marks)**

(E/P) **23** **a** Find the solution of the differential equation $16\frac{d^2y}{dx^2} + 8\frac{dy}{dx} + 5y = 5x + 23$ for which $y = 3$ and $\frac{dy}{dx} = 3$ at $x = 0$. **(8 marks)**

b Show that $y \approx x + 3$ for large values of x. **(2 marks)**

(E/P) **24** Find the solution of the differential equation $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 3\sin 3x - 2\cos 3x$ for which $y = 1$ at $x = 0$ and for which y remains finite as $x \to \infty$. **(8 marks)**

(E/P) **25** x satisfies the differential equation

$$\frac{d^2x}{dt^2} + 8\frac{dx}{dt} + 16x = \cos 4t,\ t \geqslant 0$$

a Find the general solution of the differential equation. **(8 marks)**

b Find the particular solution of this differential equation for which, at $t = 0$, $x = \frac{1}{2}$ and $\frac{dx}{dt} = 0$. **(5 marks)**

c Describe the behaviour of the function for large values of t. **(2 marks)**

Challenge

1 Use the substitution $z = y^2$ to transform the differential equation

$$2(1 + x^2)\frac{dy}{dx} + 2xy = \frac{1}{y}$$

into a differential equation in z and x. By first solving the transformed equation,

a find the general solution of the original equation, giving y in terms of x.

b Find the particular solution for which $y = 2$ when $x = 0$.

2 **a** Find the general solution of the differential equation

$$x^2\frac{d^2y}{dx^2} + 4x\frac{dy}{dx} + 2y = \ln x, \qquad x > 0,$$

using the substitution $x = e^u$, where u is a function of x.

b Find the equation of the solution curve passing through the point (1, 1) with gradient 1.

Summary of key points

1 You can solve a first-order differential equation of the form $\mathbf{\frac{dy}{dx} + P(x)y = Q(x)}$ by multiplying every term by the **integrating factor** $e^{\int P(x)dx}$.

2 The natures of the roots α and β of the **auxiliary equation** determine the **general solution** to the second-order differential equation $\mathbf{a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + c = 0.}$

You need to consider three different cases:

- **Case 1:** $\boldsymbol{b^2 > 4ac}$

  The auxiliary equation has two real roots α and β ($\alpha \neq \beta$). The general solution will be of the form $y = Ae^{\alpha x} + Be^{\beta x}$ where A and B are arbitrary constants.

- **Case 2:** $\boldsymbol{b^2 = 4ac}$

  The auxiliary equation has one repeated root α. The general solution will be of the form $y = (A + Bx)e^{\alpha x}$ where A and B are arbitrary constants.

- **Case 3:** $\boldsymbol{b^2 < 4ac}$

  The auxiliary equation has two complex conjugate roots α and β equal to $p \pm qi$. The general solution will be of the form $y = e^{px}(A\cos qx + B\sin qx)$ where A and B are arbitrary constants.

3 A **particular integral** is a function which satisfies the original differential equation.

4 To find the general solution to the differential equation $\mathbf{a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = f(x)}$,

- Solve the corresponding homogeneous equation $a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$ to find the complementary function, C.F.
- Choose an appropriate form for the particular integral, P.I., and substitute into the original equation to find the values of any coefficients.
- The general solution is y = C.F. + P.I.